



A cognitive diagnosis model for disengaged behaviors

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Accepted: 3 June 2025
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Abstract

Cognitive diagnosis assessments are frequently used for formative purposes. Due to the low-stakes nature of these assessments, students may exhibit disengaged behaviors, such as rapid guessing and item omissions. Most existing studies in cognitive diagnosis models assume that item responses are reflections of students' proficiency without considering their engagement levels. This study proposes a disengaged behavior cognitive diagnosis model (DB-CDM) that accounts for both disengaged and engaged behaviors simultaneously. We examined the performance of the DB-CDM through simulation and empirical studies. The simulation showed that the item parameters of the DB-CDM were recovered well, especially when the sample size was large and the proportion of disengaged students was small. The DB-CDM can also accurately identify disengaged students, even under some unfavorable conditions involving a large number of disengaged students. By comparing DB-CDM with the compensatory reparameterized unified model in terms of attribute classifications, we observed that the DB-CDM yielded similar if not higher attribute classifications. In the real data analysis, we found that engaged students had a lower probability of omission and guessing and a higher probability of exhibiting solution behavior compared to disengaged students. This paper provides some initial evidence to support the use of DB-CDM when disengaged behaviors occur.

Keywords Cognitive diagnosis model · Disengaged behaviors · Solution behavior · Rapid guessing · Missing responses

Introduction

Over the years, researchers and educators have used cognitive diagnosis models (CDMs) to examine students' mastery and nonmastery of a set of attributes or skills. The use of CDMs promotes personalized learning and instruction by offering educators useful insights into the cognitive profile of each student. Further, CDMs assist educators in modify-

ing instructions to match the unique learning needs of each student by identifying specific areas of strengths and weaknesses.

CDMs are often used alongside diagnostic assessments for formative purposes, which are usually considered low-stakes. In low-stakes assessments, it is not uncommon for students to exhibit disengaged behaviors (Ulitzsch et al., 2020a). Disengaged behaviors occur when students respond to items without putting forth an adequate amount of effort (Kuang and Sahin, 2023) or carrying out the essential mental operations needed to solve the problem (Arslan et al., 2021). When disengaged, students may respond to items by rapid guessing or providing no response (Ulitzsch et al., 2020a). Disengaged behaviors such as rapid guessing and item omission can lead to several negative consequences, including construct-irrelevant variance, which can impact the psychometric properties of the assessments. For example, rapid guessing and item omission may inflate the reliability of test scores and spuriously reduce the convergent validity of the assessments (Wise and DeMars, 2010). Inferences from such unreliable scores may threaten the validity and reliability of test scores, bias parameter estimates, and may not be a true representation of students' cognitive profiles (Wise and

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DeMars, 2010; Goldhammer et al., 2017). However, when students are not disengaged, they exhibit engaged or solution behaviors by exerting a sufficient amount of effort to generate responses that reflect their performance and involvement in the assessment (Wise, 2019b). Engaged test-taking behaviors enhance the measurement properties of the assessments, allowing for accurate measurements of students' knowledge, skills, and progress.

To mitigate the impact of disengaged behaviors on measurement, researchers have proposed a variety of methods. The first line of research focuses on detecting disengaged behaviors. In particular, some early approaches define a response time threshold needed for a student to provide an engaged response to distinguish disengaged and solution behaviors (Kong et al., 2007; Wise and Kong, 2005). In contrast, some recent methods use both response time and response accuracy or behaviors to set the threshold to improve the detection accuracy (Guo et al., 2016; Lee and Jia, 2014; Sahin and Colvin, 2020). Person-fit analysis has also been used to detect rapid guess or aberrant behaviors (Van der Linden and Guo, 2008). Although various approaches have been developed to detect disengaged behaviors, the guidance on the next step is lacking (Hong et al., 2020). Motivation filtering, or removing students who are considered disengaged, is a popular handling method (Wise and DeMars, 2005). Removing disengaged cases will reduce the influence of disengagement on item parameter estimation, but the reduction of sample size may lead to more uncertainty in parameter estimation. Alternatively, Hong and Cheng (2019) considered incorporating the disengaged information into item parameter estimation and proposed a robust maximum likelihood method to improve item parameter estimation for IRT models.

Another line of research is to model disengaged and engaged behaviors simultaneously, with most work being conducted in the context of item response theory (IRT). For example, Leventhal and Pastor (2024) illustrate that the item response tree (IRTree) model can be used to adjust students' ability for rapid guessing behavior, whereas Wang and Xu (2015) model rapid guessing within the mixture hierarchical modeling framework and consider both students' responses and response time in their model. Ulitzsch et al. (2020a) extend Wang and Xu's method by assuming item omission depends on students' ability and speed level. In the CDM context, Hsu et al. (2020) develop a response and response time joint model to handle rapid guessing, but it ignores item omission.

While through simulation studies, those models have been shown to produce less biased parameter estimates when disengaged behaviors occur (Hsu et al., 2020; Ulitzsch et al., 2020a; Wang and Xu, 2015), they are not without limitations. First, the majority of those models consider both item responses and response time. However, response time is not

always available in many test settings, and thus the applicability of those models is constrained. Second, many of those models only consider rapid guessing and ignore item omission, and thus may not be able to capture a common type of disengaged behavior in assessments.

To address these issues, we proposed a novel tree-based CDM to jointly model solution behavior, rapid guessing, and item omission. The goal of this study is to systematically evaluate the performance of this proposed model using both simulation and empirical studies.

Disengaged behaviors

Previous studies have shown that disengaged behaviors can have a negative impact on item and person parameter recovery, which can result in misclassification of students' diagnostic profiles and compromise the validity and reliability of test scores. Most of these studies focused on disengagement in the context of rapid guessing behaviors without considering students who exhibit item omission. This study considered both rapid guessing behavior and item omission as common behaviors students exhibit when disengaged. These behaviors have a differential impact on measurement. For example, rapid guessing behavior inflates test scores and undermines students' true ability estimation as they respond without fully engaging with the item (Rios and Deng, 2022). In contrast, item omission, where students leave items unanswered, if scored as incorrect, reduces test scores and potentially underestimates students' true ability (Xiao and Bulut, 2020). Ignoring these behaviors can bias parameter estimates and reduce classifications, compromising the measurement properties of the assessments (Rose et al., 2010; Wang and Xu, 2015).

Rapid guessing

Rapid guessing occurs when students attempt to answer as quickly as possible without solving the question (Ulitzsch et al., 2020a; Wang and Xu, 2015; Wise and Gao, 2017). The measurement properties of observed item responses from rapid guessing behavior are different from those of solution behavior, suggesting that item responses from rapid guessing behavior are unrelated to the construct measured by the test (Hsu et al., 2020; Rose et al., 2010; Wang and Xu, 2015). When overlooked, rapid guessing can bias both item and person parameters (Pokropek, 2016; Rios et al., 2017; Wang and Xu, 2015). To overcome these challenges, researchers have proposed various detection and post hoc handling approaches as well as model-based approaches (e.g., Leventhal and Pastor, 2024; Lindner et al., 2019; Wise, 2019a).

Detection and post hoc handling

Response time scoring is widely used for detecting rapid guessing behavior in assessments (Ulitzsch et al., 2020a; Wise and Kong, 2005). The response time scoring method assumes that students who answer an item too fast or below a particular threshold exhibit rapid guessing behavior. There are various methods for determining these thresholds. In most cases, the minimum amount of time required to provide an engaged response is used as a shared threshold across all items (Wise, 2019a). Other well-known methods include item-specific thresholds, where the threshold can be the average amount of time it takes a student to answer an item (Wise and Ma, 2012) or through a graphical representation of bimodal response time distribution (Wang and Xu, 2015). One of the major challenges with the response time approach is that it imposes a strong assumption that all students have the same cognitive processing abilities (Lindner et al., 2019; Rios and Deng, 2021; Soland et al., 2021). In practice, students may have different speeds, and failing to consider this can result in misclassification of rapid guessing behavior.

Researchers employ various post hoc handling approaches after rapid guessing behavior is detected (Jung and Lee, 2025; Svetina Valdivia et al., 2023). Commonly used methods include no corrective action taken after detecting rapid guessing responses, treating rapid guessing responses as incorrect, removing students with rapid guessing responses (motivation filtering) and treating rapid guessing responses as missing, followed by imputations (Jung and Lee, 2025). These different handling methods have distinct implications and consequences. For instance, retaining rapid guessing responses without adjustment can introduce measurement errors, potentially biasing item and person parameter estimates. Removing students with rapid guessing responses generally results in higher standard errors but lower bias because it potentially improves the quality of data, but reduces the sample size. Treating responses as missing and applying imputation methods introduces assumptions about the missingness mechanism, which can affect the validity of inferences if those assumptions are not met.

Model-based approach

Model-based approaches often rely on tree-based or hierarchical models for handling rapid guessing and solution behavior at the same time. Researchers have found that model-based approaches lead to accurate estimates of item and person parameters (Leventhal and Pastor, 2024; Rose et al., 2010; Ulitzsch et al., 2020a; Wang and Xu, 2015). For example, Wang and Xu (2015) used a hierarchical model that accounts for both response time and accuracy to examine rapid guessing and solution behaviors. They defined rapid guessing behavior using item-specific guessing probabilities

and modeled the solution behavior using the three-parameter logistic IRT model. Their findings showed that the hierarchical model produced unbiased item and person parameter estimates – unlike the model that accounts for solution behavior only.

Item omission

Item omission occurs when students skip items. While omissions may be caused by low proficiency, they may also be related to a lack of motivation (Wise and Gao, 2017). To appropriately handle omitted items, it is necessary to either investigate whether the omissions are due to disengagement or make an assumption about their underlying mechanism. To this end, researchers have developed model-based and non-model-based methods.

Non-model-based approach

Although item omissions are observable, their causes are not. When low proficiency is assumed to be the cause, item omissions are usually treated as incorrect or partially correct (Ulitzsch et al., 2020b). For example, in the Programme for International Student Assessment (PISA), Trends in International Mathematics and Science Study (TIMSS), and Progress in International Reading Literacy Study (PIRLS), item omissions are treated as wrong (Rose et al., 2010; Sachse et al., 2019). Another approach is to assess whether an item omission is due to disengagement using measures such as response time. The key distinction lies in the behavior preceding the omission: engaged students may skip an item after spending sufficient time considering it, while disengaged students tend to omit items with little to no effort or deliberation (Ulitzsch et al., 2020a). For example, the Program for the International Assessment of Adult Competencies (PIAAC) uses this strategy by employing a 5-s rule in the computer-based tests such that item omissions are treated as wrong if a student spends at least 5 s on an item (OECD, 2013; Sachse et al., 2019). Last, when determining the nature of omission is challenging, researchers sometimes treat them as missing values and handle them in the missing data framework. For instance, Dai and Svetina Valdivia (2022) examined the performance of various imputation methods and full-information maximum likelihood method, among others, in handling missing data for cognitive diagnosis models.

Model-based approach

Previous studies on model-based approaches integrate missing data mechanisms into IRT models to address non-ignorable item omissions (Holman and Glas, 2005; Pohl et al., 2019; Ulitzsch et al., 2020a). For example, Holman and Glas (2005) proposed a model-based approach for handling

non-ignorable missing data using IRT models. This approach allows for the estimation of the probability of an item being missing and the latent trait of individuals. Through simulations, the authors note that the model-based approach that accounts for missing data produces less biased item parameters than the models that ignore it. Pohl et al. (2019); Ulitzsch et al. (2020a) extended the model-based approach to include process data such as response time in mixture IRT models, providing a better understanding of the different cognitive processes students undertake to omit or generate a response. A major limitation of these studies is that the authors did not consider situations in which some engaged students may omit an item after spending sufficient time on it, or disengaged students put in effort to generate a response. Additionally, Holman and Glas (2005); Pohl et al. (2019); Ulitzsch et al. (2020a) used IRT models, which do not provide detailed diagnostic information about the cognitive process of students.

The proposed model

In the present study, we propose a disengaged behavior cognitive diagnosis model (DB-CDM) to analyze item response data when some students exhibit disengaged behaviors. The DB-CDM can be represented as a tree-based model as shown in Fig. 1 where circles or nodes represent latent tasks and the squares represent observed responses. In this model, a series of latent tasks are assumed when students answer test items. Latent tasks are cognitive processes that a student undertakes. The DB-CDM assumes that the first latent task for a student when approaching item j , denoted by E_j , is to decide how to answer it. Students can choose to omit it ($E_j = 0$), make a rapid guess ($E_j = 1$) or make an effort to solve the problem ($E_j = 2$). Students who choose to omit the item will produce a missing response, whereas those who guess the answer or solve the problem will have two possible outcomes (i.e., 0 for incorrect response and 1 for correct response).

Let τ represent the engagement level for a student, who is engaged if $\tau = 1$ and disengaged if $\tau = 0$. It is reasonable to assume τ affects the response of student i to latent task E of item j . In this paper, we adopt a multinomial model by assuming $P(E_j = h | \tau) = \pi_{jh}(\tau)$, with the constraints that $\pi_{jh}(\tau) \geq 0$ and $\sum_{h=0}^2 \pi_{jh}(\tau) = 1$. If the outcome of latent task E of item j is rapid guessing (i.e., $E_j = 1$), the probability of success is defined as

$$P(G_j = 1 | E_j = 1) = \frac{\exp(\gamma_j)}{1 + \exp(\gamma_j)}, \quad (1)$$

where γ_j is the guessing parameter.

If the outcome of latent task E of item j is a solution behavior (i.e., $E_j = 2$), CDMs can be used to link the rela-

tion between item responses and students' attribute profiles. Although all existing CDMs for binary data can be used, in this study, we adopted the compensatory reparameterized unified model (CRUM; Maris, 1999), a special case of the general diagnostic model (von Davier, 2005, 2019) because it has been widely used in practice (Chen and de la Torre, 2014; de la Torre, 2011; Dong et al., 2021; Ma et al., 2016). The CRUM assumes that mastering each of the required attributes increases the probability of success on an item independently of the contribution of the other attributes. Let α be a vector of attributes and by adopting the notation proposed by de la Torre (2011), we assume the first K_j^* attributes are required

for item j , where $K_j^* = \sum_{k=1}^K q_{jk}$. The CRUM is defined as

$$P(A_j = 1 | E_j = 2, \alpha) = \frac{\exp\left(\gamma_j + \sum_{k=1}^{K_j^*} \delta_{jk} \alpha_{lk}\right)}{1 + \exp\left(\gamma_j + \sum_{k=1}^{K_j^*} \delta_{jk} \alpha_{lk}\right)}, \quad (2)$$

where δ_{jk} is the main-effect terms indicating the effect of mastering attribute k . In this study, we assume the success probability due to rapid guessing is equal to the probability of success for students who do not master any attributes.

Based on the above specifications, we can define the item response function (IRF) of the DB-CDM. In particular, for item j , let X_j represent the observed response of a student with attribute pattern α_c and engagement level τ . The IRF of the DB-CDM can be obtained based on the tree structure as in

$$P(X_j = 1 | \alpha_c, \tau) = P(A_j = 1 | E_j = 2, \alpha_c)P(E_j = 2|\tau) + P(G_j = 1 | E_j = 1, \alpha_c)P(E_j = 1|\tau), \quad (3)$$

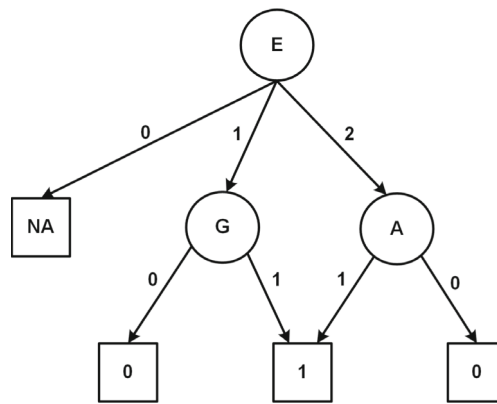
$$P(X_j = 0 | \alpha_c, \tau) = P(A_j = 0 | E_j = 2, \alpha_c)P(E_j = 2|\tau) + P(G_j = 0 | E_j = 1, \alpha_c)P(E_j = 1|\tau), \quad (4)$$

and

$$P(X_j = \text{NA} | \alpha_c, \tau) = 1 - [P(X_j = 1 | \alpha_c, \tau) + P(X_j = 0 | \alpha_c, \tau)]. \quad (5)$$

Parameter estimation

We used the marginal maximum likelihood estimation via expectation maximization (MMLE/EM) for item parameter estimation (Dempster et al., 1977). By local independence



Note. E: Response decision; G: Rapid guessing; A: Solution behavior; NA: missing response

Fig. 1 Tree diagram of DB-CDM. E: Response decision; G: Rapid guessing; A: Solution behavior; NA: missing response

assumption, the probability of observing response vector X_i for student i , given attribute pattern α_c and engagement level τ can be written as

$$P(X_i | \alpha_c, \tau) = \prod_{j=1}^J \prod_x P(X_{ij} = x | \alpha_c, \tau)^{I(X_{ij}=x)} \quad (6)$$

where $P(X_{ij} = x | \alpha_c, \tau)$ is defined in Eqs. 3 to 5, $I(X_{ij} = x)$ is an indicator function that produces a value of 1 when $X_{ij} = x$ and 0 otherwise. The EM consists of the expectation step (E-step), and maximization step (M-step). The E-step calculates the expected number of students with attribute pattern α_c and engagement level τ scoring in category x of item j . That is

$$\bar{r}_{jx} = \sum_i I(X_{ij} = x) P(\alpha_c, \tau | X_i), \quad (7)$$

such that $P(\alpha_c, \tau | X_i)$ is the posterior probability of student i with engagement level τ and attribute pattern α_c and can be calculated as

$$P(\alpha_c, \tau | X_i) = \frac{P(X_i | \alpha_c, \tau) P(\alpha_c, \tau)}{P(X_i)}. \quad (8)$$

The M-step is implemented for each item and the objective function

$$\ell = \sum_j \sum_x r_{jx} \log P(X_i | \alpha_c, \tau) \quad (9)$$

is maximized with respect to item parameters using an iterative optimization routine with a quasi-Newton method such as the limited-memory Broyden–Fletcher–Goldfarb–Shanno with bound constraints.

Simulation study

Design

A simulation study was conducted to evaluate the performance of the DB-CDM using both manipulated and fixed factors derived from previous studies (Bulut, 2021; Ma et al., 2020; Sessoms and Henson, 2018; Wise et al., 2006). Three factors, including sample size, attribute correlation, and proportion of disengaged students, were manipulated. The sample sizes considered were $N = 1000, 2000$ and 4000 , representing moderate to large samples. In empirical CDM analysis, Sessoms and Henson (2018) found that the correlation between attributes vary substantively. Ma et al. (2020) note that the correlation between attributes ranges from 0.07 to 0.95. Consistent with these studies, we considered three levels of attribute correlations, namely, 0.3, 0.5, and 0.8 representing low, moderate, and high attribute correlations, respectively. According to Bulut (2021), the levels of disengagement in large-scale assessments such as PISA 2015 ranged from 8.93% to 58.82% across several countries. Similarly, Wise et al. (2006) note that 11% to 53% of students were disengaged in five different assessments. Therefore, the proportion of disengaged students considered in this study are 20%, 40%, and 60%. It should be noted that engaged students may exhibit disengaged behaviors for some items and disengaged students may show engaged behaviors occasionally. We assume engaged students have a 15% chance to exhibit disengaged behaviors on each item, and disengaged students have a 15% chance to make effort to solve problems (see Table 1).

In the present study, test length and the number of attributes were fixed at $K = 5$ and $J = 30$, respectively, with a corresponding Q-matrix displayed in Table 2. To generate item parameters for the CRUM, we first draw $P(A_j = 0 | E_j = 2, \alpha)$ and $1 - P(A_j = 0 | E_j = 2, \alpha)$

Table 1 Parameter $\pi_{jh}(\tau)$ in the simulation study

h	Engaged students ($\tau = 1$)	Disengaged students ($\tau = 0$)
0	0.074	0.426
1	0.074	0.426
2	0.852	0.148

from $U(0.05, 0.35)$. Equation 2 is then used to generate all δ parameters, with monotonic constraints. In general, we had a total of 3 (sample size) $\times$ 3 (attribute correlation) $\times$ 3 (proportion of disengagement) = 27 conditions and 300 datasets were simulated for each condition.

Table 2 Q-matrix for simulation study

item	α_1	α_2	α_3	α_4	α_5
1	1	0	0	0	0
2	0	1	0	0	0
3	0	0	1	0	0
4	0	0	0	1	0
5	0	0	0	0	1
6	1	0	0	0	0
7	0	1	0	0	0
8	0	0	1	0	0
9	0	0	0	1	0
10	0	0	0	0	1
11	1	1	0	0	0
12	1	0	1	0	0
13	1	0	0	1	0
14	1	0	0	0	1
15	0	1	1	0	0
16	0	1	0	1	0
17	0	1	0	0	1
18	0	0	1	1	0
19	0	0	1	0	1
20	0	0	0	1	1
21	1	1	1	0	0
22	1	1	0	1	0
23	1	1	0	0	1
24	1	0	1	1	0
25	1	0	1	0	1
26	1	0	0	1	1
27	0	1	1	1	0
28	0	1	1	0	1
29	0	1	0	1	1
30	0	0	1	1	1

Criteria

We used bias and root mean square error (RMSE) to examine item parameter recovery. The bias and RMSE were calculated by

$$\text{Bias} = \frac{\sum_{r=1}^R \sum_{j=1}^J (\hat{\xi}_{rj} - \xi_{rj})}{R \times J} \quad (10)$$

and

$$\text{RMSE} = \sqrt{\frac{\sum_{r=1}^R \sum_{j=1}^J (\hat{\xi}_{rj} - \xi_{rj})^2}{R \times J}}, \quad (11)$$

where R represents the number of replications (i.e., $R = 300$) and J is the test length (i.e., $J = 30$), $\hat{\xi}_{rj}$ and ξ_{rj} are, respectively, the estimated and true parameters in each replication r .

The proportion of correctly classified attribute vectors (PCV) and the proportion of correctly identifying engagement level (PCE) were used as benchmarks for evaluating person parameter recovery. They were defined as;

$$\text{PCV} = \frac{\sum_{r=1}^R \sum_{i=1}^N I_{[\hat{\alpha}_i = \alpha_i]}^{(r)}}{R \times N}, \quad (12)$$

$$\text{PCE} = \frac{\sum_{r=1}^R \sum_{i=1}^N I_{[\hat{\tau}_i = \tau_i]}^{(r)}}{R \times N}, \quad (13)$$

where $I_{[\hat{\alpha}_i = \alpha_i]}^{(r)}$ is an indicator that returns a value of 1 when the estimated attribute vector corresponds to the true attribute vector in the r th replication and 0 otherwise, $I_{[\hat{\tau}_i = \tau_i]}^{(r)}$ equals 1 when disengaged level for student i is correctly estimated, and 0 otherwise.

Results

Item parameter recovery

Table 3 presents the mean bias and RMSE of item parameters across levels of sample sizes, attribute correlations, and the proportion of disengaged students. Several observations were made. The mean bias of rapid guessing (γ) ranges from -0.028 to -0.036 as sample size increased from 1000 to 4000, suggesting minor deviations of the $\hat{\gamma}$ from the true γ parameter for larger samples. The mean bias decreased as the correlation between the attributes increased. For example, when the attribute correlation was low ($r = 0.3$), the mean bias was -0.053, whereas when the correlation was high ($r = 0.8$), the mean bias was -0.021. Similarly, the mean bias for rapid guessing behavior was lowest when 40% of students exhibited disengaged behavior. The mean bias for π was less than

Table 3 Mean bias and RMSE of item parameters for DB-CDM

		bias			RMSE		
		π	γ	δ	π	γ	δ
<i>N</i>	1000	0.000	-0.028	0.543	0.107	0.321	1.580
	2000	0.000	-0.033	0.389	0.090	0.215	1.205
	4000	0.000	-0.036	0.339	0.083	0.132	1.004
<i>r</i>	0.3	0.000	-0.053	0.346	0.092	0.278	1.084
	0.5	0.000	-0.023	0.386	0.093	0.210	1.226
	0.8	0.000	-0.021	0.539	0.095	0.180	1.480
<i>p</i>	20	0.000	-0.045	0.409	0.099	0.184	1.301
	40	0.000	-0.012	0.402	0.095	0.196	1.173
	60	0.000	-0.039	0.461	0.087	0.288	1.316

Note: *N* = sample size, *p* = proportion of disengaged students, *r* = attribute correlation, π = engagement parameters, γ = guessing, δ = main effect

.001 under all conditions. For δ parameters, the mean bias decreased by 20.4% when the sample size increased from *N* = 1000 to 4000. Additionally, lower attribute correlation was associated with lower mean bias. Mean bias increased marginally as the number of disengaged students increased. In general, as sample size increases, item parameter recovery improves; however, increasing the proportion of disengaged students will reduce the number of effective sample sizes for students who exhibit effort behaviors, and as a result, affect the estimation accuracy of item parameters.

In the case of RMSE, larger sample sizes are associated with lower RMSEs for the π , γ , and δ parameters. For example, the RMSE for π , γ , and δ decreased by 0.024, 0.189, and 0.576 as sample sizes increased from 1000 to 4000, respectively. There was no significant difference in RMSE for π across levels of attribute correlation. The RMSE for γ decreased by 0.100 whereas that of δ increased by 0.400 when the correlation between attributes increased from 0.3 to 0.8. The RMSE for π , γ and δ increased by 0.012, 0.104, and 0.015, respectively, as the proportion of disengaged students increased from 20% to 60%. Generally, a larger sample size, lower correlation between attributes and lower proportion of disengaged students leads to higher item parameter recovery.

Attribute pattern classification

Table 4 shows the proportion of correctly classified attribute vector (PCV) for the DB-CDM and CRUM when students exhibit engaged and disengaged behaviors. From the table, the DB-CDM produced similar, if not higher, attribute pattern classification accuracy than the CRUM. In particular, the differences in attribute pattern classification accuracy were apparent for disengaged students when the proportion of disengaged students was high and attributes were strongly

Table 4 PCV of DB-CDM and CRUM for engaged and disengaged students

N	p	r	Engaged students		Disengaged students	
			DB-CDM	CRUM	DB-CDM	CRUM
1000	20	0.3	0.896	0.894	0.568	0.579
		0.5	0.912	0.910	0.561	0.598
		0.8	0.934	0.934	0.589	0.604
	40	0.3	0.892	0.884	0.562	0.551
		0.5	0.906	0.903	0.569	0.555
		0.8	0.933	0.932	0.598	0.563
	60	0.3	0.891	0.870	0.565	0.536
		0.5	0.909	0.885	0.577	0.538
		0.8	0.930	0.929	0.614	0.555
2000	20	0.3	0.903	0.901	0.590	0.584
		0.5	0.915	0.915	0.592	0.585
		0.8	0.938	0.937	0.616	0.594
	40	0.3	0.901	0.899	0.593	0.575
		0.5	0.912	0.910	0.601	0.573
		0.8	0.931	0.932	0.609	0.577
	60	0.3	0.896	0.880	0.610	0.541
		0.5	0.913	0.897	0.614	0.543
		0.8	0.929	0.927	0.643	0.554
4000	20	0.3	0.899	0.900	0.618	0.597
		0.5	0.909	0.910	0.617	0.602
		0.8	0.933	0.934	0.624	0.607
	40	0.3	0.899	0.897	0.624	0.574
		0.5	0.910	0.907	0.638	0.581
		0.8	0.933	0.931	0.649	0.590
	60	0.3	0.898	0.884	0.645	0.546
		0.5	0.910	0.895	0.665	0.547
		0.8	0.931	0.927	0.669	0.555

Note: PVC = proportion of correctly classified attribute vector, DB-CDM = Disengaged behavior CDM, CRUM = compensatory reparameterized unified model, *N* = sample size, *p* = proportion of disengaged students, *r* = attribute correlation

correlated. For example, when the sample size, attribute correlation, and proportion of disengaged students was 4000, 0.8, and 60%, respectively, the PCV for disengaged students was 0.669 for DB-CDM and 0.555 for CRUM. This finding suggests about 21% improvement in attribute pattern classification accuracy by using DB-CDM. Similarly, the DB-CDM improved by 11% when *N* = 1000, *p* = 60% and *r* = 0.8 and by 16% when *N* = 2000, *p* = 60% and *r* = 0.8. Additionally, both DB-CDM and CRUM had better attribute pattern classification for engaged students. This is not unexpected because engaged students are more likely to make an effort to solve problems, and their responses are more predictable.

Table 5 Proportion of correctly identifying engagement level for DB-CDM

<i>N</i>	<i>p</i>	<i>r</i>	Disengaged students	Engaged students
1000	20%	0.3	0.997	0.991
		0.5	0.997	0.998
		0.8	0.998	1.000
	40%	0.3	0.995	0.996
		0.5	0.997	0.994
		0.8	0.997	0.997
	60%	0.3	0.991	0.994
		0.5	0.993	0.994
		0.8	0.993	0.995
2000	20%	0.3	0.999	0.979
		0.5	1.000	0.982
		0.8	0.999	0.984
	40%	0.3	0.997	0.991
		0.5	0.996	0.991
		0.8	0.997	0.991
	60%	0.3	0.997	0.995
		0.5	0.994	0.995
		0.8	0.992	0.997
4000	20%	0.3	0.998	0.990
		0.5	0.998	0.990
		0.8	0.997	0.990
	40%	0.3	0.994	0.994
		0.5	0.994	0.995
		0.8	0.995	0.992
	60%	0.3	0.991	0.997
		0.5	0.991	0.997
		0.8	0.989	0.996

Note: *N* = sample size, *p* = proportion of disengaged students, *r* = attribute correlation

Engagement/disengagement classification

Table 5 represents the proportion of correctly identifying engagement level (PCE). It can be observed that the PCEs

were consistently above 98% across all simulation conditions. This suggests that the DB-CDM can accurately classify both engaged and disengaged students even when conditions are unfavorable such as a sample size of 1000, 60% of students are disengaged and a low correlation between attributes.

Empirical study

Data and Q-matrix

For illustration, we analyze a set of data from PISA 2015 computer-based mathematics assessments for Australia. The data consist of 1831 students' responses to nine multiple-choice items. For these items, an incorrect, correct, and no responses were coded as 0, 1, and 9, respectively. Table 6 shows the summary information of item responses. For example, 1467 students answered item 1 correctly whereas 349 and 15 students provided incorrect and missing responses, respectively. Similarly, items 7 and 8 have the highest proportion of incorrect (69.47%) and missing responses (13%), respectively.

The data were previously analyzed by Shan and Wang (2020) and they developed the Q-matrix, which involves four attributes: (α_1) change and relationship, (α_2) quantity (α_3) space and shape and (α_4) uncertainty and data. The Q-matrix is given in Table 7.

Analysis

The data were analyzed using the DB-CDM. The model fit for the DB-CDM was evaluated using Pearson $\chi^2 = \sum \frac{(f_O - f_E)^2}{f_E}$ (Agresti, 2012), where f_O and f_E denote the observed and expected frequencies, respectively. The nonparametric bootstrap approach was used to create the reference distribution for Pearson χ^2 statistic as the bootstrap method has been shown to be robust, especially in circumstances where traditional model data fit indices are sensitive

Table 6 Item-level number and proportion of incorrect, correct, and omission

Item no.	Item ID	Incorrect	Correct	Missing
1	CM033Q01S	349(19.061%)	1467(80.12%)	15(0.819%)
2	CM474Q01S	604(32.987%)	1220(66.63%)	7(0.382%)
3	CM155Q01S	483(26.379%)	1216(66.412%)	132(7.209%)
4	CM155Q04S	755(41.234%)	1030(56.253%)	46(2.512%)
5	CM411Q01S	855(46.696%)	830(45.33%)	146(7.974%)
6	CM411Q02S	944(51.557%)	819(44.73%)	68(3.714%)
7	CM803Q01S	1272(69.47%)	465(25.396%)	94(5.134%)
8	CM442Q02S	1123(61.333%)	470(25.669%)	238(12.998%)
9	CM034Q01S	1081(59.039%)	638(34.844%)	112(6.117%)

Table 7 Q-matrix for PISA 2015 computer-based mathematics assessment

Item No.	α_1	α_2	α_3	α_4
1	0	1	0	0
2	0	0	1	0
3	1	0	0	0
4	1	0	0	0
5	0	0	1	0
6	0	0	0	1
7	0	0	0	1
8	0	0	1	0
9	0	1	0	0

to minor deviations from the assumed model (Austin and Tu, 2004; Grønneberg and Foldnes, 2019). Specifically, we first computed item-level χ^2 statistics for the observed data. Then, using a nonparametric bootstrap procedure, we generated 1000 bootstrapped samples of the same size as the observed data but with replacement. We fit the DB-CDM to each of the 1000 bootstrap samples and the resulting χ^2 statistics for all items were computed. The 95th percentile of each item's χ^2 statistics from the bootstrap sample was compared to the χ^2 statistic from the observed data. An item is said to fit data adequately if its χ^2 statistic from the original data is less than the 95th percentile.

Results

Results from the analysis show that all items have good model data fit. Table 8 shows the probability of omission, rapid guessing, or solution behavior for engaged and disengaged students. For those students who are engaged, they usually have very low probabilities of omission (<.08) and high probabilities of exhibiting solution behavior (>.6). The probability of rapid guessing is generally low (<.3), with Items 2 and 7 as

two exceptions. For those students who are disengaged, they are more likely to skip or guess an item rather than make an effort. The exceptions are Items 1 and 2, where disengaged students have a chance of .816 and .555 to exhibit solution behaviors, respectively.

Discussion

CDMs have been developed to advance the use of formative assessment. This type of assessment helps teachers better understand which areas students need improvement. While these assessments are valuable, their low-stakes nature often leads to disengaged behaviors among students, such as low motivation or effort. When disengaged behaviors occur, traditional CDMs may be less effective in identifying students' strengths and weaknesses. To address this issue, this paper proposes a DB-CDM to account for both engaged and disengaged behaviors related to item omission and rapid guessing.

We examined the performance of the DB-CDM through simulations and empirical studies. The results from the simulation study show that the parameters of the DB-CDM can be recovered accurately in general. When compared with the conventional CDM (i.e., CRUM) that does not account for disengaged behaviors, the DB-CDM produces better attribute pattern classifications when the sample size or the proportion of disengaged students was relatively large. While the conventional CDMs may perform as well as the DB-CDMs in classifying engaged students, the DB-CDM tends to have a better attribute pattern classification accuracy for those students who are less engaged. Given that in practice it is unknown which students are disengaged, the DB-CDM tends to be a safer option when disengaged behaviors are suspected to occur. These findings are consistent with previous studies on disengaged behaviors in which accurate item and person parameters were generally obtained for models that accounted for disengaged behaviors when they are present

Table 8 Probabilities of items at the engagement node

Item No.	Engaged students			Disengaged students		
	Chance of omission	Chance of rapid guessing	Chance of solution behavior	Chance of omission	Chance of rapid guessing	Chance of solution behavior
1	0.002	0.171	0.828	0.064	0.120	0.816
2	0.001	0.396	0.603	0.029	0.421	0.551
3	0.035	0.210	0.755	0.405	0.177	0.418
4	0.001	0.235	0.765	0.247	0.566	0.187
5	0.027	0.182	0.791	0.557	0.315	0.128
6	0.002	0.257	0.741	0.358	0.450	0.192
7	0.009	0.312	0.680	0.434	0.409	0.157
8	0.074	0.170	0.756	0.628	0.303	0.070
9	0.023	0.293	0.684	0.406	0.472	0.122

(Ulitzsch et al., 2020a; Wang and Xu, 2015; Leventhal and Pastor, 2024; Rios et al., 2017). It should be noted that, however, the DB-CDM is complicated and thus may require a larger sample size for stable estimation of model parameters than the conventional CDMs. Additionally, the empirical study demonstrates that engaged students are more likely to engage in solution behavior, as opposed to guessing or omitting items. In contrast, disengaged students omit or guess at items rather than engage in solution behavior.

The current study has major practical and theoretical implications for understanding and modeling test-taking behaviors. Generally, our findings corroborate studies on disengagement in which students may evaluate the cognitive demands imposed by the item and make decisions on whether to omit, guess, or engage in solution behavior (Ulitzsch et al., 2020a; Nagy et al., 2023). Such non-effortful response patterns, if not handled appropriately, may inflate test score reliability and decrease the score's convergent validity, which compromises the interpretation of results (Goldhammer et al., 2017). This study shows that modeling both engaged and disengaged behaviors simultaneously can help researchers uncover patterns of disengagement, allowing for a better understanding of item and person characteristics related to disengagement (Nagy et al., 2023).

Despite its contributions, the DB-CDM is not without limitations. First, the DB-CDM may need a relatively large sample size for parameter estimation, and thus may not be ideal for some low-stakes classroom assessments. Future studies should investigate the small sample performance of the proposed model and explore how to improve its performance if it is less satisfactory. Additionally, we assumed engaged and disengaged students have a 15% chance of exhibiting disengaged behaviors and solution behaviors on each item, respectively. Future studies can explore these behaviors using ideas from Bulut (2021) and Wise et al. (2006). The monotonicity constraints and maximum a posterior approach have been used to improve attribute pattern classification accuracy of CDMs in small samples (Ma and Jiang, 2021) and they could be potentially useful for the DB-CDM. Second, we compared the proposed model to the compensatory reparameterized unified model. Future studies can examine the performance of other reduced CDM, like the deterministic inputs, noisy, "and" gate (DINA; Junker and Sijtsma, 2001) model and deterministic inputs, noisy, "or" gate (DINO; Henson et al., 2009) model and saturated CDM, like the generalized DINA (GDINA; de la Torre, 2011) with DB-CDM. This comparison can provide nuanced information on whether traditional CDMs can handle disengaged test-taking behavior better than the DB-CDM. Third, the DB-CDM considers only item responses because they are the type of data that are most widely available. Although this improves

the applicability of the DB-CDM, the model's performance could potentially be enhanced by incorporating process data such as response time and action sequences when they are available. More importantly, the proposed DB-CDM offers a valuable foundation, potentially serving as a building block for more sophisticated models that can accommodate both process and item response data. Further, identifiability is a critical issue that needs to be addressed to make sure the parameters of a model can be uniquely estimated. Recent research has examined the conditions for global identification of CDMs (e.g., Gu and Xu, 2020; Xu and Zhang, 2016), but determining the identifiability conditions is typically challenging or even intractable for complex models (Maris and Bechger, 2009). As a result, determining the identifiability conditions is deferred to future studies. Finally, the omission considered in this paper is related to disengagement. Other types of omission, such as the omission at the end of the test due to time constraints, are not considered.

Acknowledgements The code for simulation and empirical study is available at <https://tinyurl.com/4e27aaey>

Funding This paper is based upon work supported by the National Science Foundation under Grant No. 2150601. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

Data Availability Data for the empirical study is available at <https://www.oecd.org/en/data/datasets/pisa-2015-database.html>

Declarations

Conflict of interest The authors declare no conflicts of interest.

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